Warehouse Robotics: Heterogeneous Robots

Assignment D3-V3

Under Supervision:

Professor: Fulvio Mastrogiovanni

Tutor: Omar Kashmar

Presented by:

Belal Elmikkawy (S8164102)

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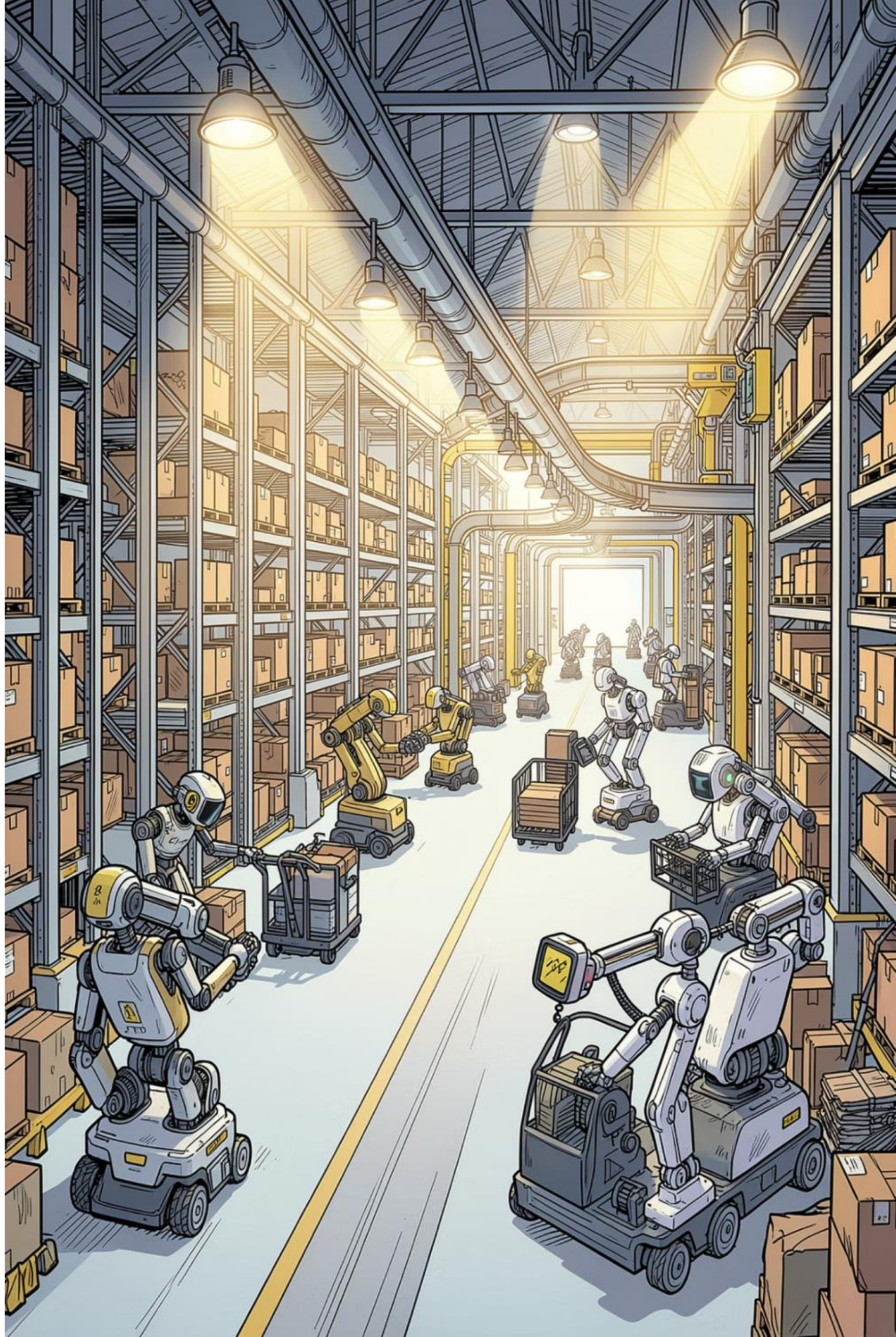


1.1 Introduction and Domain Overview

Manipulation Robots

Slow

Grippers

Cannot enter transit areas

Transport Robots

Fast

Mobile

No arms

Can navigate the entire warehouse

1

Forced Cooperation

2

Manipulation + Transport

No single robot can do both tasks, so warehouse work depends on coordination between specialized robot types.

1.2 Classical PDDL Model

1.2.1 Domain Modeling Choices

- **Typing & Predicates**: Robots are explicitly typed; state is tracked with predicates like `(manip-holding)` and `(transport-carrying)`.
- **Navigation Constraints**: `(transit-area ?l)` marks restricted zones; manipulators are kept out of transit areas.
- **Mutual Exclusion**: `(occupied ?l)` prevents two robots from occupying the same location at once.
- **Adjacent Handoffs**: Loading and unloading happen only across connected neighboring nodes.

1.2.2 Problem Scenarios – Single Robot Delivery

Baseline test: a manip-robot operates in a small area with Shelves A and B. There are no transit restrictions. The robot picks, moves, and places the package alone.

01

Pick up at Shelf A

02

Move through corridor

03

Place at Shelf B

Problem 2: Cooperative Delivery

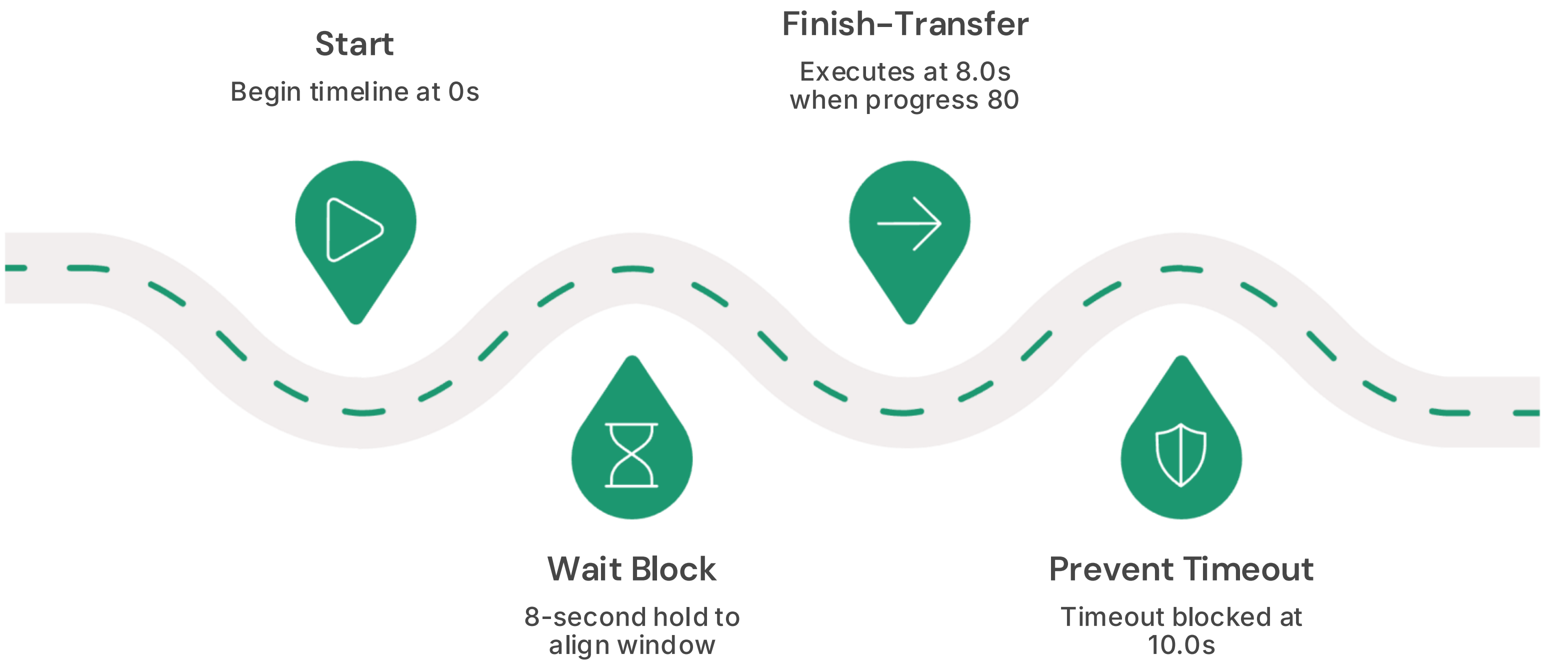
Warehouse split into zone-A and zone-C separated by a transit-area. manip1 picks up the package and loads it onto transport1. transport1 crosses the transit area. manip2 receives the package and completes delivery.

1.3 PDDL+ Continuous Time Model

Numeric fluents track completion, continuous processes advance the transfer over time, and event-triggered failures fire when progress reaches the timeout threshold.

1.4 Results and Execution

The planner inserts an **8-second wait block** to align execution with the transfer window.



This timeline shows how ENHSP schedules the action so the package is captured before the timeout event can fire.

- Planner: ENHSP (Expressive Numeric Heuristic Search Planner)
- Key Result: Planner generates 8-second wait block
- Validation: `finish-transfer` executes at exactly 8.0s when progress = 80
- Outcome: Package safely captured before timeout at 10.0s

1.5 Discussion and Limitations

Role of Heterogeneity in Planning Complexity

Specialized robot types increase the search space because each agent contributes different capabilities and constraints. At the same time, heterogeneity enables forced cooperation, since tasks must be assigned across complementary robots to achieve progress.

Abstraction of Physical Interaction

The model abstracts robot handoffs as logical state changes rather than continuous physical processes. This simplifies reasoning about coordination, but it hides geometric detail, contact dynamics, and timing-sensitive motion effects.

Limitations

- State space scalability: PDDL+ processes inflate search complexity
- Deterministic time constraints: assumes perfect timing, ignores sensor noise and latency
- Collision abstraction: topological only, missing 3D geometric constraints

Key Challenges

Deadlock prevention with mutual exclusion and timing precision in continuous processes.

Future Improvements

Integrate geometric motion planning and add stochastic uncertainty handling.

1.6 Conclusion

The integration of specialized capabilities with transit-area boundaries enables effective heterogeneous robot collaboration. The evolution from classical PDDL to PDDL+ reflects a broader shift from discrete symbolic reasoning to continuous dynamic modeling. Together, these frameworks underscore both the potential and the inherent trade-offs of advanced planning for complex robotic systems.

Thank You